Diabetes Classification

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Introduction

- Diabetes affects 38.4 million Americans (11.6% of the population)
- An additional 97.6 million adults (38%) have prediabetes
- Both numbers are expected to rise without intervention
- Project goals:
  - Explore health survey data
  - Identify risk factors and preventative behaviors
  - Build a machine learning model to predict diabetes risk
  - Help reverse the growing trend of diabetes diagnoses

Dataset

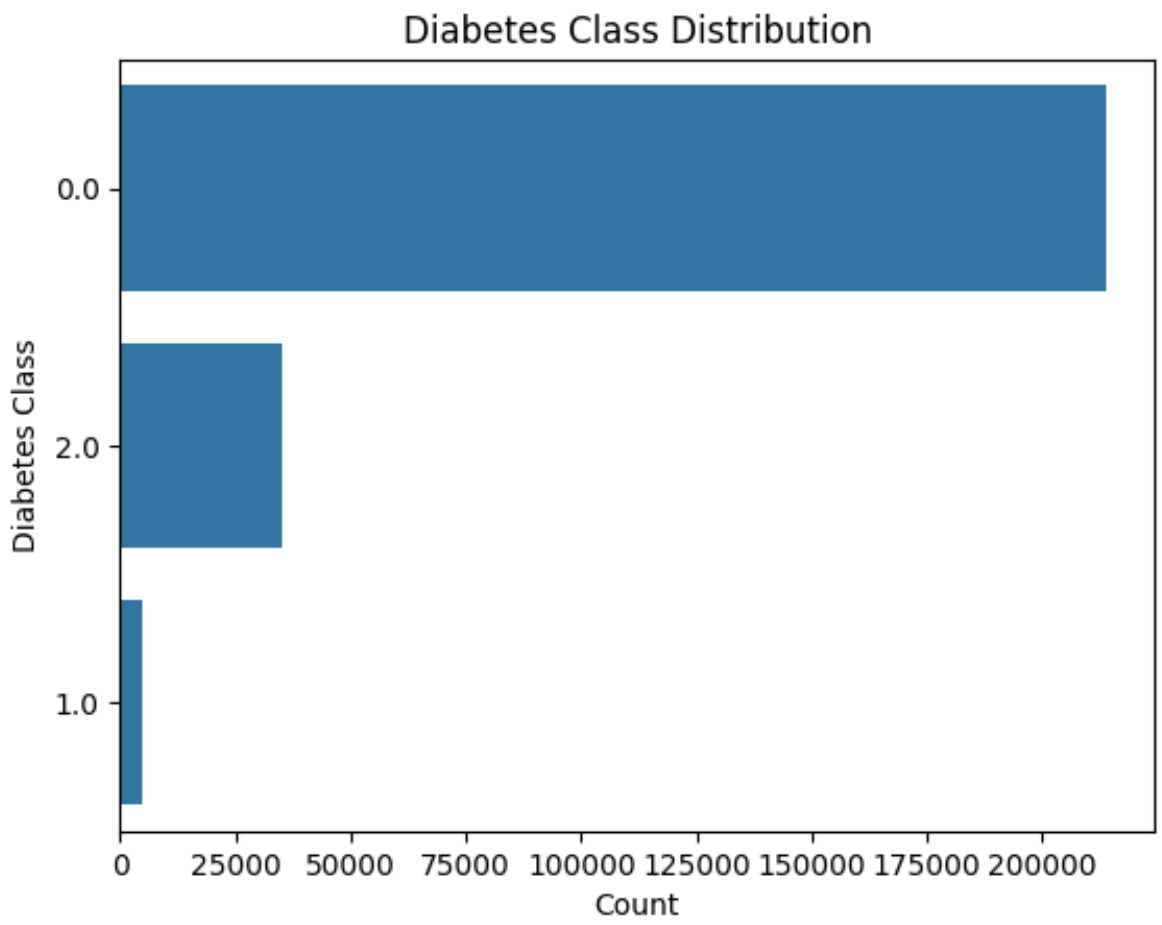
- Source: Behavioral Risk Factor Surveillance System (BRFSS) survey, 2015
- Found on Kaggle; collected via telephone interviews
- 253,680 individual survey responses
- 22 columns covering characteristics, habits, and health indicators
- Fully pre-cleaned: no missing values

Data Description

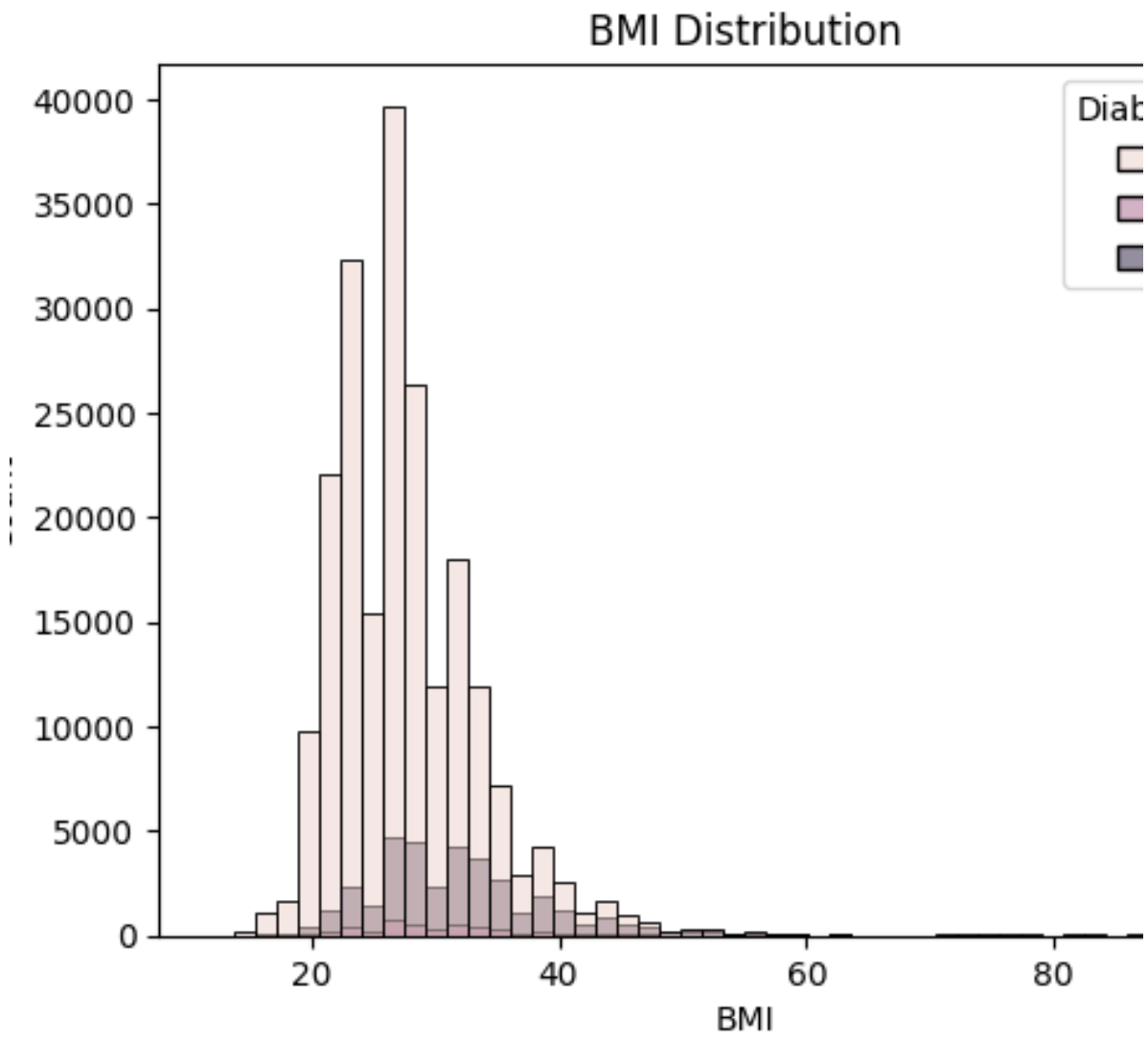
Column name	Type	Description
Diabetes_012	Categorical	0 = no diabetes, 1 = prediabetes, 2 = diabetes
HighBP	Categorical	0 = no high BP, 1 = high BP
HighChol	Categorical	0 = no high cholesterol, 1 = high cholesterol
CholCheck	Categorical	0 = no cholesterol check in 5 years, 1 = yes cholesterol check in 5 years
BMI	Numerical	Body Mass Index
Smoker	Categorical	Have you smoked at least 100 cigarettes in your entire life, 0 = no 1 = yes
Stroke	Categorical	(Ever told) you had a stroke, 0 = no 1 = yes
HeartDiseaseorAttack	Categorical	coronary heart disease or myocardial infarction, 0 = no 1 = yes
PhysActivity	Categorical	physical activity in past 30 days - not including job 0 = no 1 = yes
Fruits	Categorical	Consume Fruit 1 or more times per day 0 = no 1 = yes
Veggies	Categorical	Consume Vegetables 1 or more times per day 0 = no 1 = yes
HvyAlcoholConsump	Categorical	Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week) 0 = no 1 = yes
AnyHealthcare	Categorical	Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no 1 = yes
NoDocbcCost	Categorical	Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no 1 = yes
GenHlth	Ordinal	Would you say that in general your health is: scale 1-5 1 = excellent 2 = very good 3 = good 4 = fair 5 = poor
MentHlth	Numerical	Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? scale 1-30 days
PhysHlth	Numerical	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good? scale 1-30 days
DiffWalk	Categorical	Do you have serious difficulty walking or climbing stairs? 0 = no 1 = yes
Sex	Categorical	0 = female 1 = male
Age	Ordinal	13-level age category, 1 = 18-24, 9 = 60-64, 13 = 80 or older
Education	Ordinal	Education level scale 1-6, 1 = Never attended school or only kindergarten, 2 = Grades 1 through 8 (Elementary), 3 = Grades 9 through 11 (Some high school), 4 = Grade 12 or GED (High school graduate), 5 = College 1 year to 3 years (Some college or technical school), 6 = College 4 years or more (College graduate)
Income	Ordinal	Income scale 1-8, 1 = less than $10,000, 5 = less than $35,000, 8 = $75,000 or more

Questions of Interest

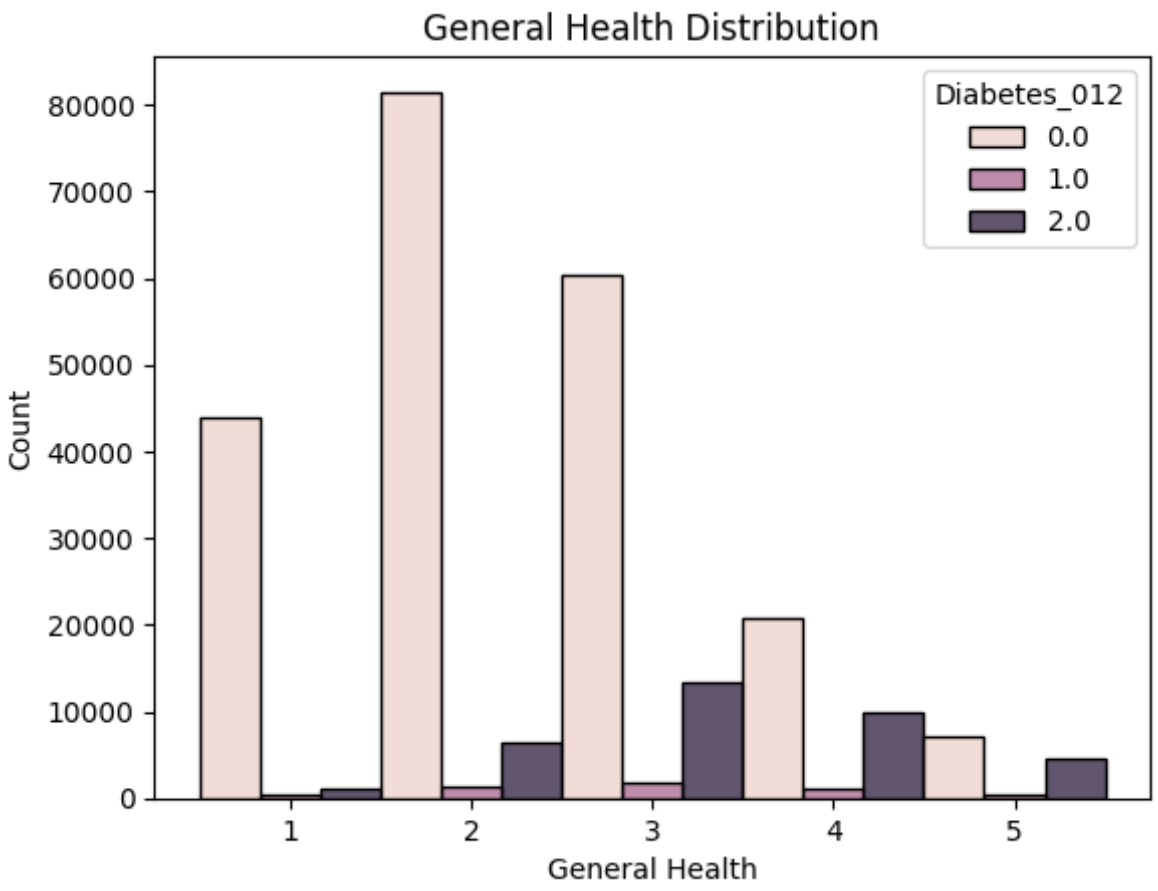
- What does the distribution of key health indicators (such as diabetes status, BMI, and general health) look like among survey respondents?
- Can we build a predictive model that accurately identifies individuals at risk for diabetes based on survey data?
- Is it more effective to predict prediabetes and diabetes separately, or to group them into a single “at risk” category?



- The majority of survey respondents reported no diabetes.
- Prediabetes and diabetes cases make up a much smaller portion of the data.



- The histogram of BMI shows a right-skewed distribution.
- Most respondents cluster between 20 and 40, with a long tail stretching toward higher values.



- Most respondents ranked their own general health as very good, good, and excellent followed by fair and poor.

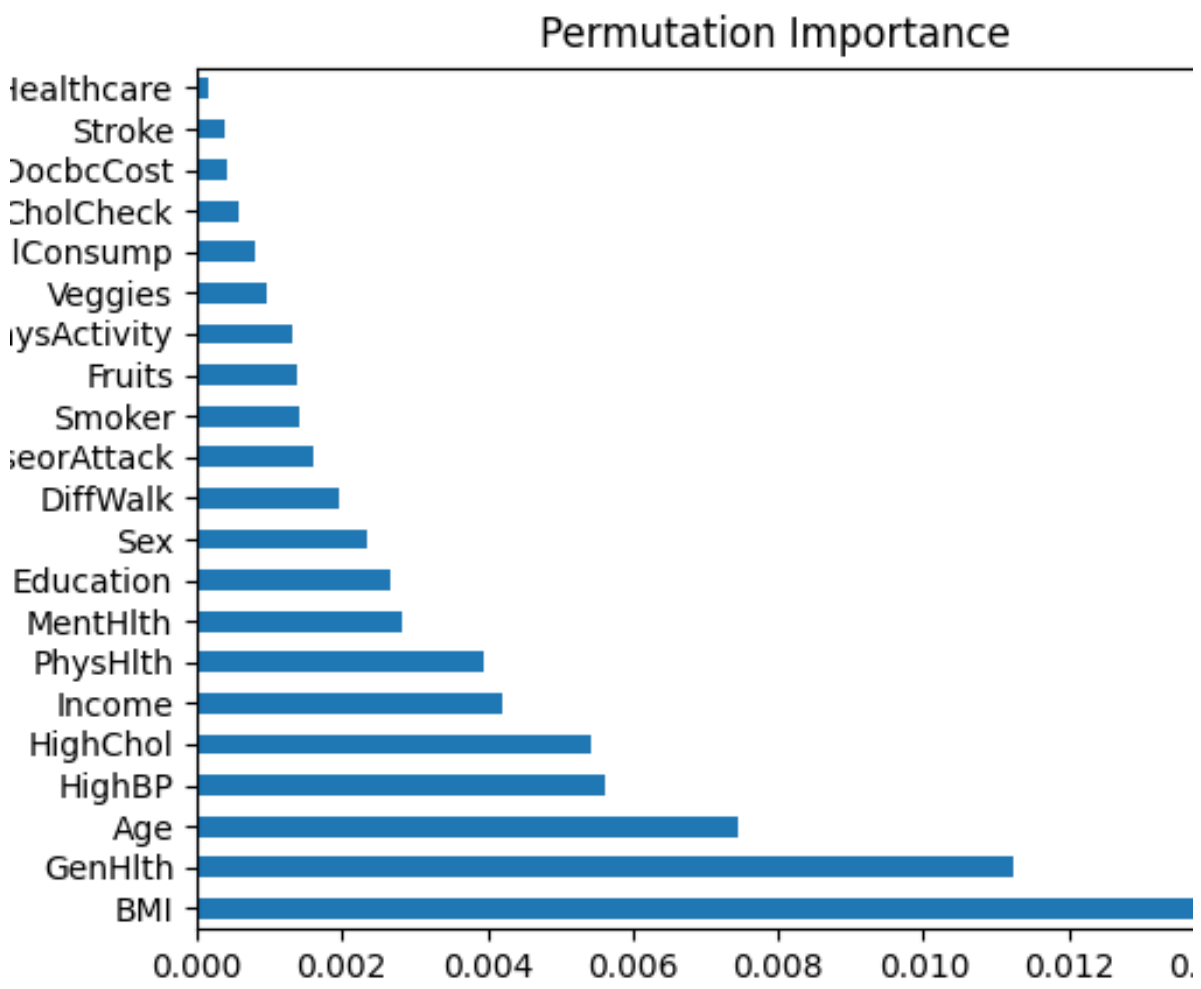
Baseline Modeling

- Used a Random Forest Classifier with F1 Score as the chosen performance metric
- Tuned the following hyperparameters within the respective boundaries
  - N estimators: 100 to 300
  - Max depth: 5 to 10
  - Min samples split: 2 to 10
  - Min samples leaf: 1 to 10
- Wanted to avoid accuracy as a performance metric due to the data being high imbalanced

Parameters (6)

Search parameters

Parameter	Value
max_depth	10
min_samples_leaf	1
min_samples_split	10
n_estimators	101
n_jobs	-1
random_state	42

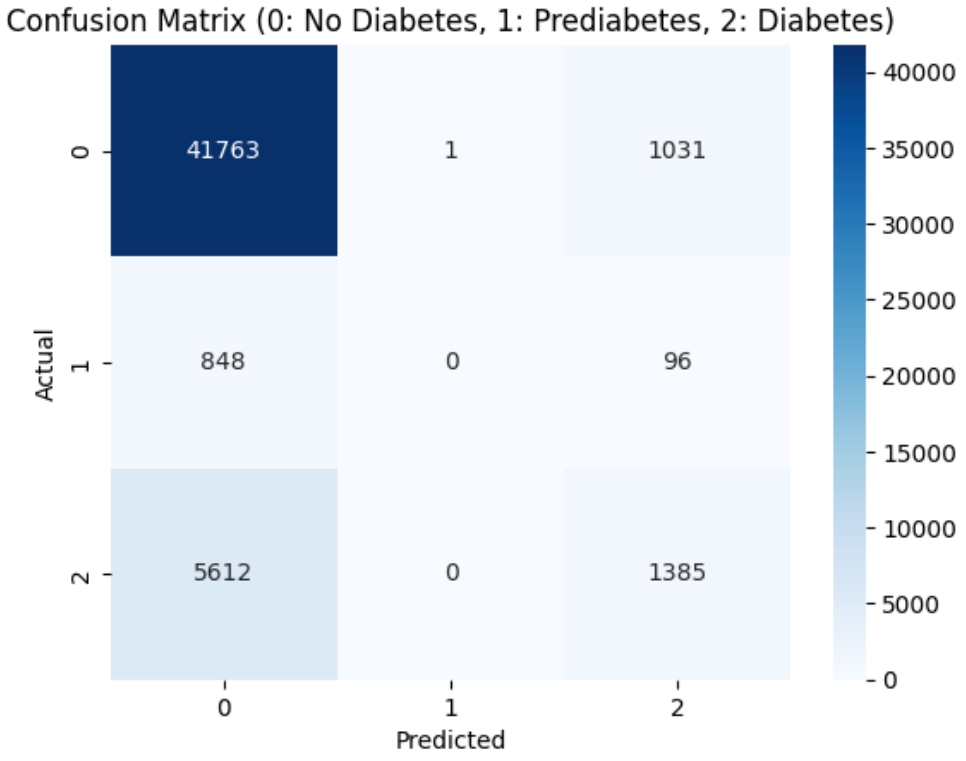


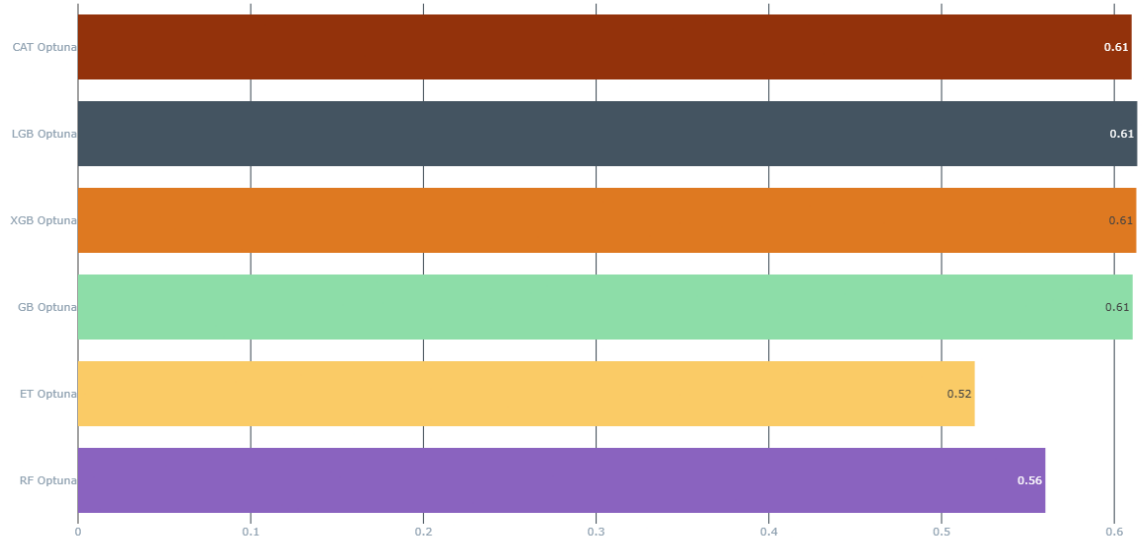
Modeling Feature Importances

- Feature engineering didn't lead to improved performance
- We used permutation importance to better understand which variables the model relied on most.
- This analysis confirmed the relevance of features like BMI, general health, age, and blood pressure.

Modeling Issues

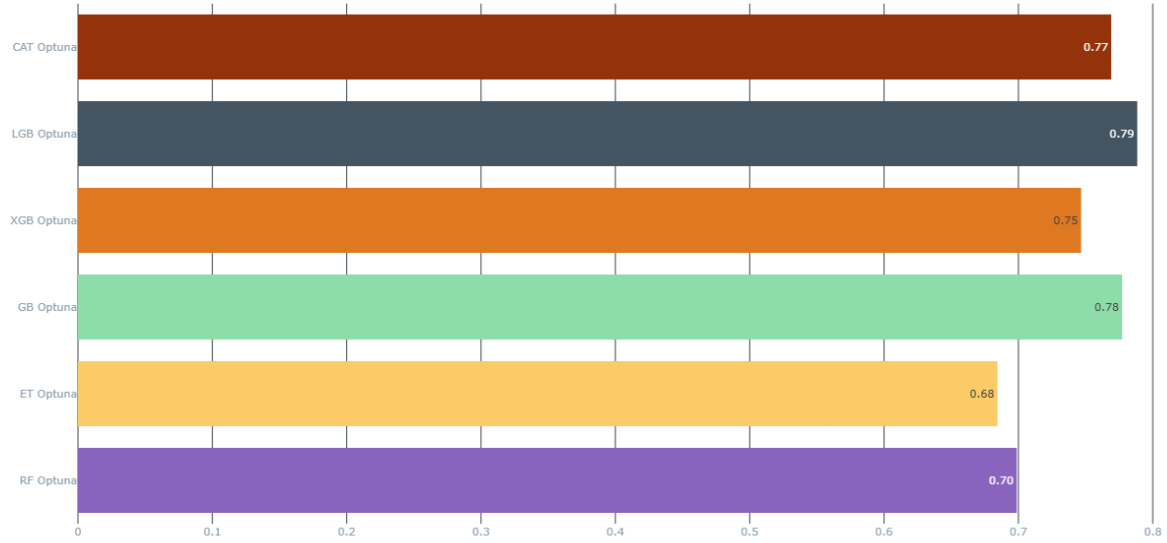
- The best observed macro-averaged F1 score hovered in the low 40s
- Challenged by the imbalanced target variable and the overlap between diabetes and prediabetes cases.
- The confusion matrix confirmed the model's poor performance on the prediabetes class, with zero correct predictions.
- All prediabetes cases were misclassified as either no diabetes or diabetes, underscoring the model's inability to distinguish this class.





Modeling Results

- Simplified the task by combining diabetes and prediabetes into a single “at risk” category for binary classification.



Modeling Optimization

- We achieved even better results by tuning the decision threshold between the "no diabetes" and "at risk" classes.

Conclusion & Further Study

- Reframing the task as binary classification (at risk vs. no diabetes) improved model performance
- Optimized LightGBM model achieved an F1 score of 78.8%
- Binary framing made clinical sense: prediabetes and diabetes share risk factors and interventions
- Model could support early intervention by identifying high-risk individuals more effectively
- Explore automated machine learning (AutoML) tools (Auto-sklearn, FLAML, H2O AutoML)